Detailed Diagnostic Report (DDR)

Professional Dilapidation and Damage Report (DDR)

Date: October 26, 2023

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1. Property Issue Summary

A comprehensive assessment utilizing thermal verification has identified several areas of concern across the property, primarily related to water ingress and structural integrity. Widespread dampness issues are noted in multiple internal living spaces, including the Hall, Bedroom, Master Bedroom, and Kitchen. The Master Bedroom exhibits efflorescence, indicative of active moisture migration. Plumbing integrity is compromised in the Common Bathroom, and significant leakage is present in the Parking area. Furthermore, the external wall shows signs of cracking, which can contribute to penetrating damp. All identified issues have undergone thermal verification, confirming the presence of anomalies consistent with moisture and thermal bridging.

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2. Area-wise Observations

The following observations were made during the property assessment:

Hall: Skirting dampness was observed. Thermal verification available.

Bedroom: Dampness detected within the bedroom area. Thermal verification available.

Master Bedroom: Pronounced wall dampness, accompanied by efflorescence (salt deposits), indicating active moisture ingress and evaporation. Thermal verification available.

Kitchen: Dampness observed within the kitchen area. Thermal verification available.

Common Bathroom: Identified issues include noticeable tile gaps and underlying plumbing deficiencies. Thermal verification available.

Parking: Significant leakage was detected in the parking area. Thermal verification available.

External wall: Cracks were identified on the external façade of the property. Thermal verification available.

Thermal Summary Observations:

Thermal verification confirmed temperature anomalies consistent with moisture presence and potential heat transfer issues across various areas. Hotspots were recorded ranging from 25.1°C to 28.8°C, while coldspots were identified between 20.1°C and 23.4°C. These variations are indicative of thermal bridges, areas of moisture evaporation, or trapped moisture within building materials.

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3. Root Cause Analysis

Based on the observations, the primary root causes for the identified issues are attributed to:

Water Ingress: The widespread dampness in the Hall, Bedroom, Master Bedroom, and Kitchen, coupled with parking leakage and external wall cracks, strongly suggests multiple sources of water ingress. This could stem from:

Penetrating Damp: Originating from external sources such as the identified external wall cracks, faulty rendering, or roof defects.

Rising Damp: Suggested by the skirting dampness in the Hall, potentially due to a compromised or absent damp-proof course.

Plumbing Leaks: Directly indicated by issues in the Common Bathroom (tile gaps, plumbing issues) and potentially contributing to dampness in adjacent areas. Parking leakage may also indicate a compromised underground pipe or slab defect.

Structural Defects: The detected cracks on the external wall could be due to building settlement, material fatigue, or differential movement. These cracks serve as direct pathways for water penetration.

Material Degradation: Efflorescence in the Master Bedroom is a secondary effect, caused by salts migrating to the surface with moisture and crystallizing as the water evaporates. This confirms the presence and movement of water within the wall structure.

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4. Severity Assessment

Dampness (Hall, Bedroom, Kitchen): Moderate. If left unaddressed, it can lead to mould growth, plaster degradation, and compromised indoor air quality.

Wall Dampness + Efflorescence (Master Bedroom): Moderate to High. Efflorescence indicates a persistent and active moisture problem that is actively damaging finishes and potentially structural elements over time.

Tile Gaps + Plumbing Issues (Common Bathroom): High. These are direct and active sources of water leakage, contributing significantly to internal dampness and potential structural damage to floors/walls if not promptly repaired.

Parking Leakage: High. Indicates a significant breach in waterproofing or a pipe leak, potentially undermining the structural integrity of the slab or foundation over time and causing considerable damage.

Cracks Detected (External wall): Moderate to High. The severity depends on the crack size and type. These are direct pathways for water ingress, leading to penetrating damp and can be indicative of underlying structural movement requiring further investigation.

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5. Recommendations

Immediate action is recommended to mitigate further damage and ensure the longevity of the property:

1. Comprehensive Leak Detection & Repair: Conduct a thorough investigation to pinpoint the exact sources of water ingress for all damp areas (Hall, Bedroom, Master Bedroom, Kitchen, Parking). This should include:

Plumbing Inspection: A specialist plumbing inspection for the Common Bathroom and surrounding areas to identify and rectify all leaks and seal tile gaps.

Roof & External Inspection: A detailed inspection of the roof, external walls, and drainage systems to identify entry points for penetrating damp.

Slab Inspection: Investigate the source of parking leakage, potentially requiring localized excavation or further thermal/moisture mapping.

2. External Wall Remediation:

Rake out and repair all detected cracks on the external wall using appropriate flexible fillers or mortar, ensuring a watertight seal.

Consider overall external façade inspection and potential re-rendering or waterproofing treatment if cracks are widespread.

3. Dampness Treatment & Remediation:

Once the source of dampness is rectified, allow all affected internal walls to dry out thoroughly. Dehumidifiers and ventilation may be required.

Remove efflorescence from the Master Bedroom wall using appropriate methods (e.g., dry brushing, specialized cleaners), followed by re-plastering and redecoration once dry.

Investigate and implement solutions for rising damp in the Hall skirting, which may include chemical damp-proof course injection or physical DPC installation.

4. Regular Monitoring: Establish a schedule for regular inspections of the property, especially after heavy rainfall, to detect new or recurring issues early.

5. Professional Consultation: Engage qualified building surveyors, waterproofing specialists, and structural engineers where necessary to provide detailed assessments and supervise

repair works.

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6. Missing Information

To provide a more comprehensive and precise report, the following information would be beneficial:

Specific Thermal Anomaly Locations: While thermal verification is confirmed, specific details linking individual hotspot/coldspot temperatures to particular defects in each room (e.g., "Hall skirting dampness corresponds to a 28.8°C hotspot") would enhance analysis.

Moisture Meter Readings: Quantifiable moisture content readings for affected materials (walls, skirting) to establish the extent and severity of dampness.

Dimensions and Nature of Cracks: Detailed description of external wall cracks (e.g., width, length, depth, active/dormant, hairline/structural) to better assess their impact and required repair method.

Property Age and Construction Type: Information on the building's age, construction materials, and foundation type can help in understanding common issues and potential causes.

History of Maintenance and Repairs: Any previous repair attempts for dampness, leaks, or cracks would provide context.

Environmental Context: Information on recent weather conditions, proximity to water bodies, or site-specific drainage issues.

Internal Environmental Conditions: Ambient temperature and relative humidity at the time of the inspection, which influence thermal readings and condensation potential.

Appendix: Extracted Images

Inspection Images

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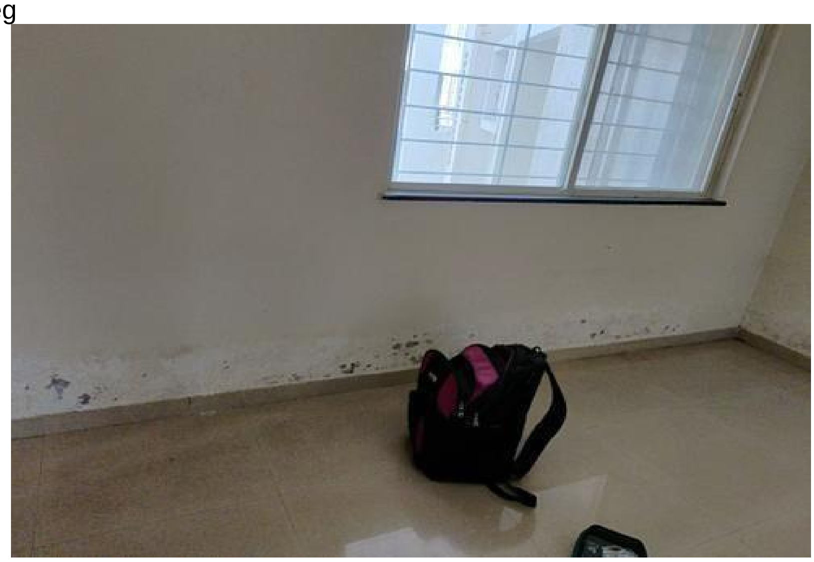

Thermal Images

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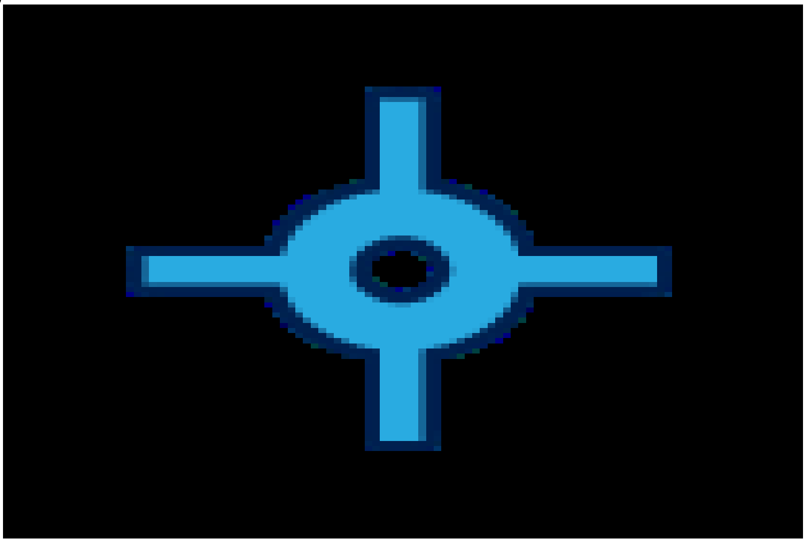

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